Computational Marine Ecological Modelling (25314)

# NPZDO model

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# List of Symbols

| Symbol | Unit | Description |
|---|---|---|
| $C$ | varies | Generic tracer concentration |
| $D$ | $\mathrm{mmol\,N\,m^{-3}}$ | Detrital organic nitrogen concentration in the water column |
| $D_{\mathrm{n}}$ | $\mathrm{mmol\,N\,m^{-3}}$ | Detritus concentration in the bottom water cell |
| $D_{\mathrm{sed}}$ | $\mathrm{mmol\,N\,m^{-2}}$ | Sediment detritus inventory |
| $F_{\mathrm{air-sea}}$ | $\mathrm{mmol\,O_2\,m^{-2}\,d^{-1}}$ | Air–sea oxygen exchange flux |
| $F_{\mathrm{ben}}$ | $\mathrm{mmol\,N\,m^{-2}\,d^{-1}}$ | Benthic remineralization flux |
| $F_{\mathrm{dep}}$ | $\mathrm{mmol\,N\,m^{-2}\,d^{-1}}$ | Deposition flux of detritus from the water column to the sediment |
| $F_{\mathrm{graz}}$ | $\mathrm{mmol\,N\,m^{-3}\,d^{-1}}$ | Grazing flux |
| $F_{\mathrm{mortP}}$ | $\mathrm{mmol\,N\,m^{-3}\,d^{-1}}$ | Phytoplankton mortality flux |
| $F_{\mathrm{mortZ}}$ | $\mathrm{mmol\,N\,m^{-3}\,d^{-1}}$ | Zooplankton mortality flux |
| $F_{\mathrm{prod}}$ | $\mathrm{mmol\,N\,m^{-3}\,d^{-1}}$ | Primary production flux |
| $F_{\mathrm{relin}}$ | $\mathrm{mmol\,N\,m^{-2}\,d^{-1}}$ | Sediment nutrient release flux |
| $F_{\mathrm{remin}}$ | $\mathrm{mmol\,N\,m^{-3}\,d^{-1}}$ | Water-column remineralization flux |
| $F_{\mathrm{resp}}$ | $\mathrm{mmol\,N\,m^{-3}\,d^{-1}}$ | Respiration and excretion flux returning nitrogen to dissolved nutrients |
| $G_{\mathrm{lim}}$ | – | Grazing limitation factor |
| $g_{\mathrm{max}}$ | $\mathrm{d^{-1}}$ | Maximum zooplankton grazing rate |
| $J$ | varies | Vertical tracer flux |
| $k_L$ | $\mathrm{\mu mol\,m^{-2}\,s^{-1}}$ | Light half-saturation constant |
| $k_N$ | $\mathrm{mmol\,N\,m^{-3}}$ | Nutrient half-saturation constant |
| $k_{N,\mathrm{n}}$ | $\mathrm{m\,d^{-1}}$ | Sediment–water nitrogen exchange velocity |
| $k_Z$ | $\mathrm{mmol\,N\,m^{-3}}$ | Grazing half-saturation constant |
| $L$ | $\mathrm{\mu mol\,m^{-2}\,s^{-1}}$ | Local light intensity |
| $L_{\mathrm{lim}}$ | – | Light limitation factor |
| $L_0$ | $\mathrm{\mu mol\,m^{-2}\,s^{-1}}$ | Surface light intensity |
| $m_P$ | $\mathrm{d^{-1}}$ | Phytoplankton mortality rate |
| $m_Z$ | $(\mathrm{mmol\,N\,m^{-3}})^{-1}\,\mathrm{d^{-1}}$ | Zooplankton mortality coefficient |
| $n$ | – | Number of vertical layers |
| $N$ | $\mathrm{mmol\,N\,m^{-3}}$ | Dissolved inorganic nitrogen concentration in the water column |
| $N_{\mathrm{lim}}$ | – | Nutrient limitation factor |

| | | |
|---|---|---|
| $N_{\mathrm{n}}$ | $\mathrm{mmol\,N\,m^{-3}}$ | Dissolved inorganic nitrogen concentration in the bottom water cell |
| $N_{\mathrm{sed}}$ | $\mathrm{mmol\,N\,m^{-2}}$ | Sediment nitrogen inventory |
| $O$ | $\mathrm{mmol\,O_2\,m^{-3}}$ | Dissolved oxygen concentration |
| $O_{\mathrm{atm}}$ | $\mathrm{mmol\,O_2\,m^{-3}}$ | Atmospheric oxygen concentration |
| $O_{\mathrm{crit}}$ | $\mathrm{mmol\,O_2\,m^{-3}}$ | Critical oxygen concentration |
| $O_{\mathrm{lim}}$ | – | Oxygen limitation factor |
| $O_{\mathrm{lim,n}}$ | – | Oxygen limitation factor evaluated in the bottom water cell |
| $P$ | $\mathrm{mmol\,N\,m^{-3}}$ | Phytoplankton biomass concentration |
| $R(C)$ | varies | Reaction term for tracer $C$ |
| $r_{\mathrm{ae}}$ | $\mathrm{d^{-1}}$ | Aerobic remineralization rate constant |
| $r_{\mathrm{an}}$ | $\mathrm{d^{-1}}$ | Anaerobic remineralization rate constant |
| $t$ | d | Time |
| $w$ | $\mathrm{m\,d^{-1}}$ | Generic vertical sinking velocity |
| $w_D$ | $\mathrm{m\,d^{-1}}$ | Detrital sinking velocity |
| $y_P$ | $\mathrm{mmol\,O_2\,mmol\,N^{-1}}$ | Stoichiometric oxygen yield associated with primary production |
| $y_N$ | $\mathrm{mmol\,O_2\,mmol\,N^{-1}}$ | Stoichiometric oxygen consumption associated with remineralization |
| $z_i$ | m | Depth at the center of grid cell $i$ |
| $z_{\mathrm{sed}}$ | m | Effective thickness of the sediment reservoir |
| $Z$ | $\mathrm{mmol\,N\,m^{-3}}$ | Zooplankton biomass concentration |
| $\Delta O$ | $\mathrm{mmol\,O_2\,m^{-3}}$ | Transition width for oxygen limitation |
| $\Delta z$ | m | Vertical grid spacing |
| $\epsilon_D$ | – | Fraction of grazing routed to detritus |
| $\epsilon_N$ | – | Fraction of grazing recycled to dissolved nutrients |
| $\kappa$ | $\mathrm{m^2\,d^{-1}}$ | Vertical diffusivity |
| $\mu$ | $\mathrm{d^{-1}}$ | Realized phytoplankton growth rate |
| $\mu_{\mathrm{max}}$ | $\mathrm{d^{-1}}$ | Maximum phytoplankton growth rate |

# Usage of Large Language Models

AI was not used in this report for research purposes. ChatGPT was only used to rephrase already written abstracts. Since none of the authors are native English speakers, the tool helped improve spelling and wording to meet an academic standard.

Large Language Models were also used to produce part of the programming of the assignment.

# Code Availability

The full model code used in this study is available on GitHub:

`https://github.com/matteo-mvh/Ecological-Modeling-Course`

# Introduction

Marine ecosystem models provide a framework for studying the interactions between biological production, nutrient cycling, and physical transport in the ocean. A commonly used simplified representation is the NPZD framework, which describes the dynamics of nutrients (N), phytoplankton (P), zooplankton (Z), and detritus (D). In this study, the classical NPZD model is extended by including dissolved oxygen (O), forming an NPZDO ecosystem model in which oxygen regulates remineralization and biological activity, particularly under low-oxygen conditions.

The model is formulated as a one-dimensional vertical water column forced by seasonal variations in light and vertical mixing. Physical transport processes are represented through vertical diffusion and the sinking of particulate organic matter, while the biological dynamics include phytoplankton growth, zooplankton grazing, mortality, detritus remineralization, and oxygen production and consumption.

The objective of this report is to present the structure of the NPZDO model, describe the governing equations and numerical implementation, and analyze the resulting ecosystem dynamics.

# 1 Model Description

## 1.1 Model Concept

The developed ecosystem model is a vertically resolved one-dimensional NPZDO model that represents the coupled evolution of nutrients, plankton biomass, detrital organic matter, and dissolved oxygen in the water column. In addition to the pelagic ecosystem, the model includes a simplified representation of sediment reservoirs that interact with the overlying water column. The framework provides a conceptual description of the main biological and biogeochemical components of a seasonally forced water column and resolves the vertical distribution of ecosystem tracers.

The water column is discretized on a vertical grid, while the sediment is represented by two vertically integrated reservoirs per unit area. This formulation captures first-order benthic recycling and enables exchanges between the sediments and the overlying water column without introducing additional vertical complexity in the sediment domain.

### 1.1.1 State Variables

The model contains five prognostic state variables within the water column and two additional sediment reservoirs:

| Symbol | State variable | Unit |
|---|---|---|
| $N$ | Dissolved inorganic nitrogen | $\mathrm{mmol\,N\,m^{-3}}$ |
| $P$ | Phytoplankton biomass | $\mathrm{mmol\,N\,m^{-3}}$ |
| $Z$ | Zooplankton biomass | $\mathrm{mmol\,N\,m^{-3}}$ |
| $D$ | Detrital organic nitrogen | $\mathrm{mmol\,N\,m^{-3}}$ |
| $O$ | Dissolved oxygen | $\mathrm{mmol\,O_2\,m^{-3}}$ |
| $D_{\mathrm{sed}}$ | Sediment detritus | $\mathrm{mmol\,N\,m^{-2}}$ |
| $N_{\mathrm{sed}}$ | Sediment nutrients | $\mathrm{mmol\,N\,m^{-2}}$ |

All biological tracers except oxygen are expressed in nitrogen units to ensure consistent mass conservation between ecosystem compartments. The sediment variables represent vertically integrated pools at the seabed and therefore do not possess vertical structure.

### 1.1.2 Conceptual Illustration

Figure 1.1 illustrates the ecosystem compartments and fluxes represented in the NPZDO model.

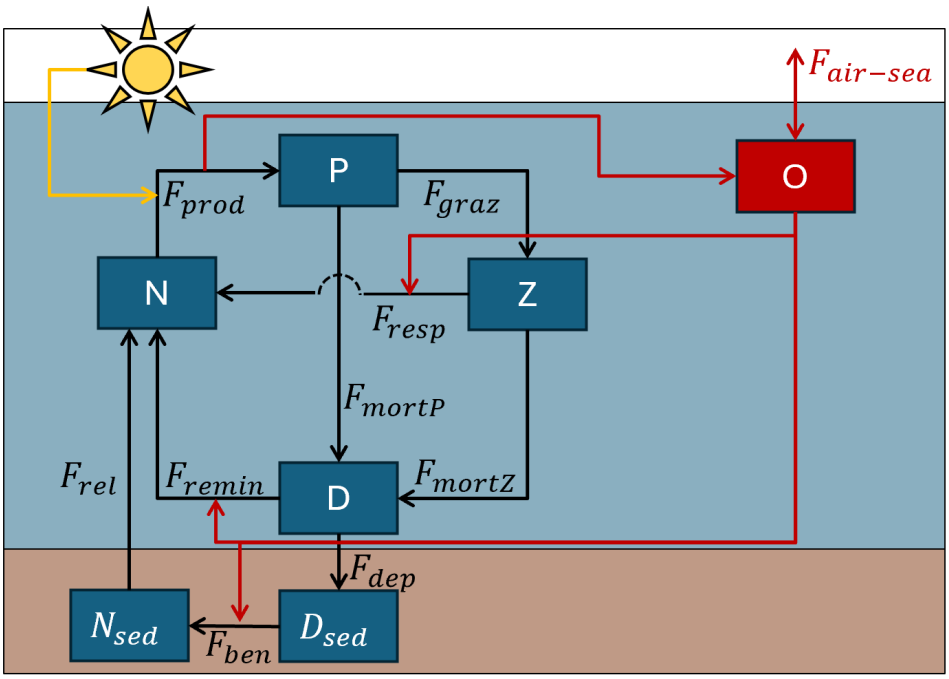


**Figure 1.1:** Schematic illustration of the NPZDO model showing the state variables and the biological, chemical, and sediment interactions represented in the model.

The biological interactions represented in the model include primary production, grazing, mortality, remineralization, sediment deposition, and benthic nutrient recycling. Oxygen is produced during photosynthesis and consumed during respiration and remineralization, while exchange with the atmosphere occurs through air–sea gas exchange.

All fluxes included in the model are summarized in Table 1.1.

| Flux symbol | Process name | Unit |
|---|---|---|
| $F_{\mathrm{prod}}$ | Primary production | $\mathrm{mmol\,N\,m^{-3}\,d^{-1}}$ |
| $F_{\mathrm{graz}}$ | Zooplankton grazing | $\mathrm{mmol\,N\,m^{-3}\,d^{-1}}$ |
| $F_{\mathrm{resp}}$ | Zooplankton respiration and excretion | $\mathrm{mmol\,N\,m^{-3}\,d^{-1}}$ |
| $F_{\mathrm{mortP}}$ | Phytoplankton mortality | $\mathrm{mmol\,N\,m^{-3}\,d^{-1}}$ |
| $F_{\mathrm{mortZ}}$ | Zooplankton mortality | $\mathrm{mmol\,N\,m^{-3}\,d^{-1}}$ |
| $F_{\mathrm{remin}}$ | Detritus remineralization | $\mathrm{mmol\,N\,m^{-3}\,d^{-1}}$ |
| $F_{\mathrm{dep}}$ | Detritus deposition to sediment | $\mathrm{mmol\,N\,m^{-2}\,d^{-1}}$ |
| $F_{\mathrm{ben}}$ | Benthic remineralization | $\mathrm{mmol\,N\,m^{-2}\,d^{-1}}$ |
| $F_{\mathrm{rel}}$ | Nutrient release from sediment | $\mathrm{mmol\,N\,m^{-2}\,d^{-1}}$ |
| $F_{\mathrm{air-sea}}$ | Air–sea oxygen exchange | $\mathrm{mmol\,O_2\,m^{-2}\,d^{-1}}$ |

**Table 1.1:** Fluxes included in the NPZDO ecosystem model.

### 1.1.3 Vertical Discretization

The water column is represented by a one-dimensional vertical grid extending from the surface to the seabed. As illustrated in Figure 1.2, the depth axis is discretized into $n$ vertically stacked computational cells.

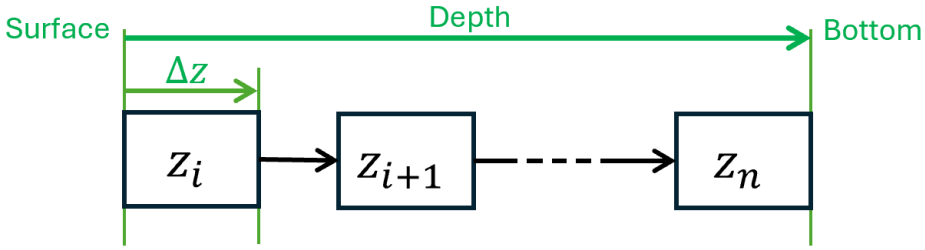


**Figure 1.2:** Schematic representation of the vertical discretization used in the model. The water column is divided into $n$ depth cells extending from the surface to the bottom. Pelagic state variables are defined in each grid cell, while sediment variables are represented as integrated reservoirs at the seabed.

The vertical grid is defined as

$$z_i = \left(i - \frac{1}{2}\right)\Delta z, \quad i = 1, 2, \ldots, n \tag{1.1}$$

where $\Delta z$ is the grid spacing and $n$ is the number of vertical layers.

All pelagic state variables are defined at the center of each grid cell, forming vertically resolved concentration profiles that evolve over time. Vertical transport between adjacent cells represents turbulent mixing and the sinking of particulate material, allowing the model to resolve vertical gradients in biological activity, nutrient availability, and oxygen concentration.

The sediment variables $N_{\text{sed}}$ and $D_{\text{sed}}$ are treated as vertically integrated reservoirs at the seabed. Exchange with the water column occurs through deposition and benthic remineralization fluxes applied at the bottom boundary. While this simplified representation captures first-order benthic recycling, a vertically resolved sediment model could provide a more detailed representation of internal sediment gradients.

## 1.2 Physical Transport

All pelagic tracers in the model are subject to vertical transport processes. The temporal evolution of a tracer concentration $C$ is therefore described by a vertical advection–diffusion–reaction equation of the form

$$\frac{\partial C}{\partial t} = -\frac{\partial J}{\partial z} + R(C), \tag{1.2}$$

where $J$ denotes the vertical tracer flux and $R(C)$ represents the reaction terms arising from biological processes.

The vertical flux of a tracer is formulated as the sum of an advective and a diffusive contribution:

$$J = wC - \kappa \frac{\partial C}{\partial z}, \tag{1.3}$$

where $w$ is the sinking velocity and $\kappa$ is the vertical diffusivity. Positive values of $w$ correspond to downward transport. Substituting this flux definition into the general conservation equation gives

$$\frac{\partial C}{\partial t} = -\frac{\partial}{\partial z}\left(wC - \kappa \frac{\partial C}{\partial z}\right) + R(C). \tag{1.4}$$

In the present model, sinking is prescribed for phytoplankton and detritus, while nutrients, zooplankton, and oxygen do not experience vertical advection. Diffusivity varies seasonally and with depth, thereby representing changes in upper-ocean mixing intensity throughout the year.

## 1.3 Biogeochemical Processes

The biological dynamics of the NPZDO model can be expressed in terms of the fluxes illustrated in Figure 1.1. These fluxes represent the transfer of nitrogen between the ecosystem compartments and the exchange of organic matter with the sediment reservoir.

The local rate of change of the nitrogen-based state variables in the water column is given by

$$\frac{dN}{dt} = -F_{\text{prod}} + F_{\text{resp}} + F_{\text{remin}} + F_{\text{rel}}, \tag{1.5}$$

$$\frac{dP}{dt} = F_{\text{prod}} - F_{\text{graz}} - F_{\text{mortP}}, \tag{1.6}$$

$$\frac{dZ}{dt} = F_{\text{graz}} - F_{\text{mortZ}} - F_{\text{resp}}, \tag{1.7}$$

$$\frac{dD}{dt} = F_{\text{mortP}} + F_{\text{mortZ}} - F_{\text{remin}} - F_{\text{dep}}. \tag{1.8}$$

The temporal evolution of the sediment nitrogen and detritus reservoirs is described by

$$\frac{dD_{\text{sed}}}{dt} = F_{\text{dep}} - F_{\text{ben}}, \tag{1.9}$$

$$\frac{dN_{\text{sed}}}{dt} = F_{\text{ben}} - F_{\text{rel}}. \tag{1.10}$$

The sign convention of the individual flux terms follows directly from the conceptual illustration in Figure 1.1. Fluxes entering a compartment contribute positively to its rate of change, while fluxes leaving a compartment appear with a negative sign. In this way, the governing equations directly reflect the direction of the flux arrows shown in the schematic representation and ensure conservation of nitrogen between the modeled compartments.

Oxygen is included as an additional tracer in the model and is coupled to the biological processes through production during photosynthesis and consumption during respiration and remineralization. In addition, oxygen exchanges with the atmosphere through an air–sea flux at the surface boundary.

### 1.3.1 Phytoplankton Growth

Primary production corresponds to the flux from dissolved nutrients to phytoplankton biomass and is denoted by $F_{\text{prod}}$ in Figure 1.1. The production flux is defined as

$$F_{\text{prod}} = \mu P, \tag{1.11}$$

where $\mu$ is the realized phytoplankton growth rate. The growth rate is limited by both light availability and nutrient supply and is given by

$$\mu = \mu_{\text{max}} \min(L_{\text{lim}}, N_{\text{lim}}), \tag{1.12}$$

where $\mu_{\text{max}}$ is the maximum phytoplankton growth rate, $L_{\text{lim}}$ is the light limitation factor, and $N_{\text{lim}}$ is the nutrient limitation factor.

The limitation terms are expressed as Monod-type functions:

$$L_{\text{lim}} = \frac{L}{L + k_L}, \tag{1.13}$$

$$N_{\text{lim}} = \frac{N}{N + k_N}. \tag{1.14}$$

Here $L$ represents the local light intensity and $N$ the dissolved inorganic nitrogen concentration. The parameters $k_L$ and $k_N$ are half-saturation constants for light and nutrient uptake, respectively. Physically, these constants define the resource level at which phytoplankton growth reaches half of its maximum potential rate. Smaller values of $k_L$ or $k_N$ therefore indicate more efficient utilization of the corresponding resource.

This formulation implies that phytoplankton growth is limited by whichever of the two resources, light or nutrients, is most restrictive at a given depth and time.

### 1.3.2 Zooplankton Grazing

Zooplankton grazing represents the transfer of phytoplankton biomass to zooplankton and is denoted by the flux $F_{\text{graz}}$ in Figure 1.1. The grazing flux is parameterized as

$$F_{\text{graz}} = g_{\max} \min(G_{\text{lim}}, O_{\text{lim}}) Z, \tag{1.15}$$

where $g_{\max}$ is the maximum grazing rate, $G_{\text{lim}}$ is the prey limitation factor, and $O_{\text{lim}}$ represents the oxygen limitation of grazing.

The prey limitation is given by

$$G_{\text{lim}} = \frac{P^2}{P^2 + k_Z^2}, \tag{1.16}$$

where $P$ denotes the phytoplankton biomass and $k_Z$ is the grazing half-saturation constant. Consequently, grazing increases with phytoplankton biomass but gradually saturates at high prey concentrations. The quadratic formulation introduces a sigmoidal response that suppresses grazing at low prey concentrations while allowing saturation at high phytoplankton biomass.

In addition to prey availability, grazing is also limited by oxygen availability. To avoid an abrupt threshold behaviour under low oxygen conditions, oxygen limitation is represented by a smooth hyperbolic tangent switch:

$$O_{\text{lim}} = \frac{1}{2}\left[1 + \tanh\left(\frac{O - O_{\text{crit}}}{\Delta O}\right)\right], \tag{1.17}$$

where $O_{\text{crit}}$ is the critical oxygen concentration and $\Delta O$ determines the sharpness of the transition. This formulation allows grazing activity to gradually decline as oxygen concentrations decrease, representing reduced metabolic activity under hypoxic conditions.

The grazed phytoplankton biomass is not entirely converted into zooplankton biomass. Instead, the grazing flux is partitioned into three fractions to ensure mass conservation:

- a fraction $\epsilon_N$ is immediately recycled back into dissolved nutrients, representing rapid recycling through processes such as excretion and metabolic losses,
- a fraction $\epsilon_D$ is routed into the detrital pool, representing unassimilated organic matter produced during grazing,
- the remaining fraction $(1-\epsilon_N-\epsilon_D)$ contributes to zooplankton growth and represents the assimilated portion of the grazing flux.

This partitioning ensures that the full grazing flux is redistributed among the ecosystem compartments while conserving nitrogen mass.

### 1.3.3 Mortality Fluxes

In addition to grazing, both phytoplankton and zooplankton experience mortality losses. These losses contribute to the detrital pool and are represented by the mortality fluxes

$$F_{\mathrm{mortP}} = m_P P, \tag{1.18}$$

$$F_{\mathrm{mortZ}} = m_Z Z^2. \tag{1.19}$$

Here, $m_P$ is the phytoplankton mortality parameter and $m_Z$ is the zooplankton mortality parameter. Phytoplankton mortality is modeled as a linear loss process, while zooplankton mortality follows a quadratic formulation. The quadratic dependence represents density-dependent losses such as predation or crowding effects that increase at higher zooplankton concentrations.

### 1.3.4 Water-Column Remineralization

Detrital organic matter in the water column is remineralized back into dissolved inorganic nitrogen through the combined remineralization flux $F_{\mathrm{remin}}$. In the present model formulation, this flux consists of an aerobic and an anaerobic contribution acting simultaneously:

$$F_{\mathrm{remin}} = \underbrace{r_{\mathrm{ae}}\, O_{\mathrm{lim}}\, D}_{\text{aerobic}} + \underbrace{r_{\mathrm{an}}\, D}_{\text{anaerobic}} \ . \tag{1.20}$$

Here, $D$ is the detritus concentration, $r_{\mathrm{ae}}$ and $r_{\mathrm{an}}$ are the aerobic and anaerobic remineralization rate constants, and $O_{\mathrm{lim}}$ is the oxygen limitation factor introduced in Equation 1.17.

This formulation represents the simultaneous action of oxygen-dependent and oxygen-

independent remineralization pathways. The aerobic contribution decreases under low-oxygen conditions through the factor $O_{\text{lim}}$, while the anaerobic contribution remains active independently of oxygen availability. Consequently, total remineralization is reduced under oxygen-poor conditions but does not vanish completely. Only the aerobic fraction contributes to oxygen consumption and therefore acts as a sink of dissolved oxygen in the water column.

### 1.3.5 Reaction Terms

Based on the process formulations introduced in the previous section, the local biogeochemical source and sink terms for the vertically resolved nitrogen tracers can be written as

$$\frac{dN}{dt} = \underbrace{-\mu P}_{\text{uptake}} + \underbrace{\epsilon_N g_{\max} \min(G_{\text{lim}}, O_{\text{lim}}) Z}_{\text{recycling}} + \underbrace{(r_{\text{ae}} O_{\text{lim}} D + r_{\text{an}} D)}_{\text{remineralization}}, \tag{1.21}$$

$$\frac{dP}{dt} = \underbrace{\mu P}_{\text{production}} - \underbrace{g_{\max} \min(G_{\text{lim}}, O_{\text{lim}}) Z}_{\text{grazing}} - \underbrace{m_P P}_{\text{mortality}}, \tag{1.22}$$

$$\frac{dZ}{dt} = \underbrace{(1 - \epsilon_N - \epsilon_D)\, g_{\max} \min(G_{\text{lim}}, O_{\text{lim}}) Z}_{\text{growth}} - \underbrace{m_Z Z^2}_{\text{mortality}}, \tag{1.23}$$

$$\frac{dD}{dt} = \underbrace{m_P P}_{\text{P mortality}} + \underbrace{\epsilon_D g_{\max} \min(G_{\text{lim}}, O_{\text{lim}}) Z}_{\text{egestion}} + \underbrace{m_Z Z^2}_{\text{Z mortality}} - \underbrace{(r_{\text{ae}} O_{\text{lim}} D + r_{\text{an}} D)}_{\text{remineralization}}. \tag{1.24}$$

Together with the vertical transport formulation introduced in Section 1.2, these reaction terms define the full set of governing equations used in the numerical model. The only remaining components required to close the system are the boundary conditions, which are described in the following sections.

## 1.4 Sediment Dynamics

The interaction between the water column and the seabed is represented through exchanges of organic matter and dissolved nutrients at the sediment–water interface. In the present formulation, detrital material reaching the bottom water cell is transferred into a vertically integrated sediment reservoir. Only detritus and dissolved inorganic nitrogen are exchanged with the sediment, while the remaining tracers are subject to zero-flux boundary conditions at the seabed.

### 1.4.1 Sediment Tendency

Within the sediment, deposited detritus is remineralized and converted into dissolved inorganic nitrogen. This benthic recycling represents the primary internal process governing the evolution of the sediment reservoirs.

The remineralization of sediment detritus is described by

$$F_{\mathrm{ben}} = r_{\mathrm{ae}} O_{\mathrm{lim,n}} D_{\mathrm{sed}} + r_{\mathrm{an}} D_{\mathrm{sed}}, \tag{1.25}$$

where $D_{\mathrm{sed}}$ is the sediment detritus inventory and $O_{\mathrm{lim,n}}$ is the oxygen limitation evaluated in the bottom water. As in the water-column formulation, remineralization consists of simultaneous aerobic and anaerobic pathways, with the aerobic component decreasing under low-oxygen conditions.

The evolution of the sediment detritus pool follows from the balance between deposition of particulate organic matter and benthic remineralization:

$$\frac{dD_{\mathrm{sed}}}{dt} = \underbrace{w_D D_{\mathrm{n}}}_{\text{deposition}} - \underbrace{(r_{\mathrm{ae}} O_{\mathrm{lim,n}} D_{\mathrm{sed}} + r_{\mathrm{an}} D_{\mathrm{sed}})}_{\text{benthic remineralization}}. \tag{1.26}$$

The sediment nitrogen reservoir evolves according to

$$\frac{dN_{\mathrm{sed}}}{dt} = \underbrace{(r_{\mathrm{ae}} O_{\mathrm{lim,n}} D_{\mathrm{sed}} + r_{\mathrm{an}} D_{\mathrm{sed}})}_{\text{remineralization}} - \underbrace{k_{N,\mathrm{n}} \left( \frac{N_{\mathrm{sed}}}{z_{\mathrm{sed}}} - N_{\mathrm{n}} \right)}_{\text{exchange}}. \tag{1.27}$$

The first term represents the production of dissolved inorganic nitrogen through remineralization of sediment detritus, while the second term describes the exchange of nitrogen between the sediment reservoir and the overlying bottom water.

### 1.4.2 Nitrogen Exchange at the Boundary

The coupling between the water column and the sediment occurs through the exchange of detritus and dissolved nutrients at the sediment–water interface.

The downward transfer of detritus from the water column into the sediment is represented by the deposition flux

$$F_{\mathrm{dep}} = w_D D_{\mathrm{n}}, \tag{1.28}$$

where $w_D$ denotes the sinking velocity of detrital particles and $D_\mathrm{n}$ is the detritus concentration in the lowest water column grid cell.

In addition to organic matter deposition, dissolved inorganic nitrogen can be exchanged between the sediment reservoir and the bottom water through

$$F_\mathrm{rel} = -k_{N,\mathrm{n}} \left( \frac{N_\mathrm{sed}}{z_\mathrm{sed}} - N_\mathrm{n} \right), \tag{1.29}$$

where $k_{N,\mathrm{n}}$ is the sediment–water exchange velocity, $N_\mathrm{sed}$ is the nitrogen inventory stored in the sediment reservoir, $z_\mathrm{sed}$ is the effective sediment thickness, and $N_\mathrm{n}$ is the dissolved inorganic nitrogen concentration in the bottom water cell. According to the model sign convention, positive fluxes are directed downward into the sediment, such that negative values correspond to an upward release of nitrogen back into the water column.

## 1.5 Oxygen Dynamics

In addition to the nitrogen-based ecosystem tracers, the model explicitly represents dissolved oxygen in the water column. Oxygen dynamics are closely coupled to the biological nitrogen cycle through fixed stoichiometric relationships. The coefficients $y_P$ and $y_N$ describe the amount of oxygen produced or consumed per unit of nitrogen transformed during biological processes. In this formulation, primary production generates oxygen in proportion to phytoplankton growth, while respiration and remineralization consume oxygen during the recycling of organic nitrogen. Conceptually, this coupling can be expressed as

$$\Delta O = y_P \, \Delta N_\mathrm{prod} - y_N \, \Delta N_\mathrm{remin}, \tag{1.30}$$

where $\Delta N_\mathrm{prod}$ represents nitrogen incorporated into phytoplankton biomass during primary production and $\Delta N_\mathrm{remin}$ represents nitrogen released during respiration and remineralization processes. This simplified relationship illustrates how changes in nitrogen fluxes are translated into corresponding oxygen production and consumption within the model.

### 1.5.1 Oxygen Tendency

Within the water column, oxygen production and consumption follow directly from the biological fluxes of the ecosystem model. The oxygen tendency associated with these internal processes can be written as

$$\frac{dO}{dt} = \underbrace{y_P \mu P}_{\text{production}} - \underbrace{y_N \epsilon_N g_{\max} \min(G_{\text{lim}}, O_{\text{lim}}) Z}_{\text{respiration}} - \underbrace{y_N r_{\text{ae}} O_{\text{lim}} D}_{\text{remineralization}} . \tag{1.31}$$

The first term represents oxygen production during phytoplankton growth. The second term describes oxygen consumption associated with respiration and rapid recycling of grazed phytoplankton biomass. The third term represents oxygen consumption during aerobic remineralization of detritus within the water column.

### 1.5.2 Oxygen Exchange at the Boundaries

In addition to the internal biological processes, oxygen concentrations are influenced by exchange processes at the upper and lower boundaries of the water column.

At the air–sea interface, oxygen exchanges with the atmosphere through a restoring-type gas exchange flux,

$$F_{\text{air-sea}} = k_{\text{surf}} \left( O_{\text{atm}} - O_{\text{surf}} \right) , \tag{1.32}$$

where $k_{\text{surf}}$ is the surface gas transfer coefficient, $O_{\text{atm}}$ is the equilibrium oxygen concentration imposed by the atmosphere, and $O_{\text{surf}}$ is the oxygen concentration in the uppermost model cell. When surface waters are undersaturated relative to the atmosphere, this formulation leads to a flux of oxygen into the ocean, whereas supersaturated conditions result in oxygen outgassing.

At the seabed, oxygen is consumed through aerobic remineralization of organic matter within the sediment. Detritus that reaches the bottom water cell is transferred into the sediment reservoir and subsequently remineralized through the benthic remineralization flux. The aerobic component of this process consumes oxygen in the overlying bottom water and therefore imposes an oxygen demand at the sediment–water interface.

The resulting bottom oxygen flux can be written as

$$F_{\text{sed}} = -y_N r_{\text{ae}} O_{\text{lim,n}} D_{\text{sed}}, \tag{1.33}$$

where the negative sign indicates a flux of oxygen from the bottom water into the sediment due to aerobic remineralization. This formulation links the benthic oxygen consumption directly to the aerobic remineralization rate of the sediment detritus pool $D_{\text{sed}}$.

# 2 Model Setup and Convergence

## 2.1 Numerical Implementation

The NPZDO ecosystem model is solved on a one-dimensional vertical grid extending over the prescribed water depth. As described in Section 1.1.3, the water column is discretized into $n_z$ equally spaced grid cells, and all pelagic state variables are defined at the cell centres.

Vertical transport is formulated in flux form using the tracer flux introduced in Equation 1.3, which provides a conservative discretization of both turbulent diffusion and sinking transport. The governing equations are therefore written in a finite-volume-like form, where the divergence of the interface fluxes determines the concentration tendency in each grid cell.

After spatial discretization, the model equations form a coupled system of ordinary differential equations of the form

$$\frac{d\mathbf{y}}{dt} = f(\mathbf{y}, t), \tag{2.1}$$

where the state vector $\mathbf{y}$ contains the vertical profiles of all pelagic tracers together with the sediment reservoirs $N_{\text{sed}}$ and $D_{\text{sed}}$.

All physical forcing and biological source terms are evaluated within the model tendency function at every solver call. In particular, vertical diffusivity, surface light, biological light attenuation, and all nonlinear limitation factors are recomputed from the instantaneous model state, making the system fully nonlinear and time dependent.

Time integration is carried out using the implicit backward differentiation formula (BDF) solver available in `SciPy`. This solver is well suited for the present model because the coupled system exhibits stiff behaviour arising from the interaction of relatively fast biological reactions, diffusive transport, sediment coupling, and nonlinear oxygen-dependent source terms. The numerical integration is performed with relative and absolute tolerances of $10^{-7}$ and $10^{-10}$, respectively, while the maximum time step is limited to one day in order to resolve seasonal forcing and rapid biological transitions adequately.

## 2.2 Convergence Checks

To verify that the numerical solution is robust and independent of discretization choices, several convergence tests were performed. These include a grid convergence analysis as well as a time convergence diagnostic.

### 2.2.1 Grid Convergence

To assess the sensitivity of the numerical solution to vertical resolution, the model was integrated using different numbers of vertical layers while keeping all parameters, forcing functions, and solver settings identical. The tested grid sizes ranged from $n_z = 10$ to $n_z = 70$ layers over the 200 m water column.

For each simulation, selected diagnostics were compared against the highest-resolution simulation ($n_z = 70$), which was used as the reference solution. Differences were quantified using the relative $L_2$ error norm. For a tracer field $C$, the relative error between a simulation and the reference solution is defined as

$$E_{L_2} = \frac{\sqrt{\sum_{i=1}^{n_z} \left(C_i - C_i^{\mathrm{ref}}\right)^2}}{\sqrt{\sum_{i=1}^{n_z} \left(C_i^{\mathrm{ref}}\right)^2}}. \tag{2.2}$$

Here $C_i$ denotes the tracer value in grid cell $i$ for the tested simulation and $C_i^{\mathrm{ref}}$ is the corresponding value from the reference run.

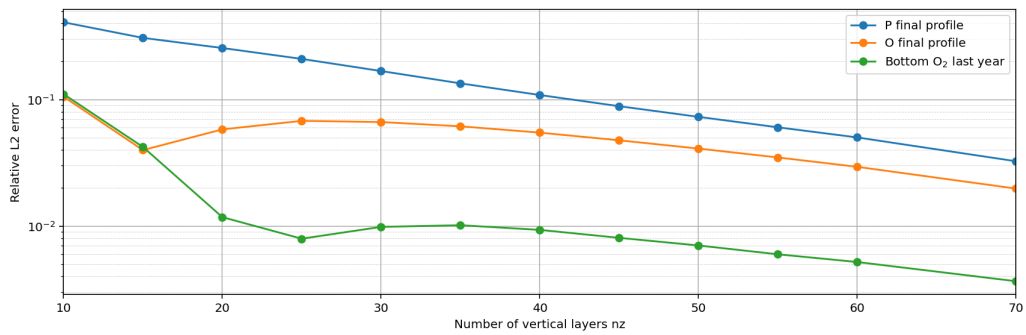


**Figure 2.1:** Grid convergence analysis showing the relative $L_2$ error of selected diagnostics compared to the highest-resolution simulation. The plotted quantities include the final phytoplankton profile, the final oxygen profile, and the bottom oxygen concentration during the final simulation year.

Figure 2.1 shows that the numerical solution approaches a grid-independent state as the vertical resolution increases. The phytoplankton profile exhibits the strongest sensitivity to grid resolution, whereas oxygen-related diagnostics converge more rapidly. The bottom oxygen concentration reaches relative errors of order $10^{-2}$ already at moderate grid

resolutions, indicating that vertically integrated or bottom-layer diagnostics are less sensitive to discretization than detailed tracer profiles.

Based on this analysis, a resolution of $n_z = 50$ layers was selected for the remaining simulations, providing a good compromise between numerical accuracy and computational efficiency.

### 2.2.2 Time Convergence

In addition to spatial convergence, the model must approach a stable repeating seasonal cycle during the spin-up phase. To evaluate this behaviour, yearly averages of selected diagnostics were computed and compared to the final simulation year using the relative error defined in Equation 2.2.

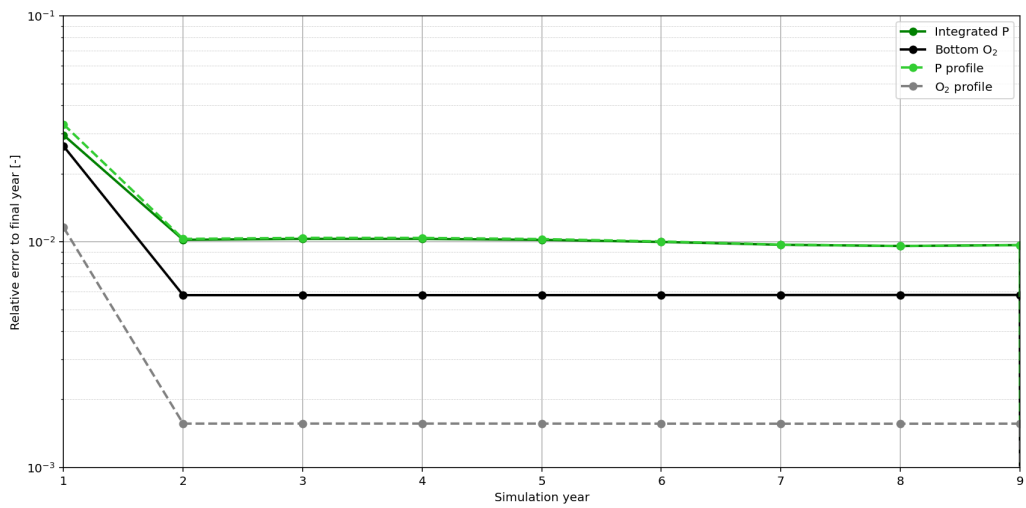


**Figure 2.2:** Spin-up convergence of selected diagnostics relative to the final simulation year. Solid lines show vertically integrated phytoplankton biomass and bottom-water oxygen, while dashed lines represent vertically resolved annual mean profiles.

As shown in Figure 2.2, the relative differences decrease rapidly during the first simulation year and remain small thereafter. The vertically integrated phytoplankton inventory stabilizes at relative differences of approximately $10^{-2}$, while oxygen-related diagnostics show slightly smaller deviations.

This rapid convergence is largely driven by the seasonal forcing. During winter, low light conditions suppress primary production and effectively reset biological activity in the water column, causing the ecosystem state to repeat quickly from year to year.

Based on this analysis, a total simulation length of four years was used for the subsequent experiments, consisting of three years of spin-up followed by one year used for analysis.

# 3 Model Behaviour and Analysis

## 3.1 Factors Limiting Phytoplankton Growth

Phytoplankton growth in the NPZDO model is controlled by resource availability and ecological interactions that regulate the realized growth rate. In the present formulation, the primary environmental constraints on phytoplankton production are light availability and dissolved inorganic nitrogen.

Light determines the energy available for photosynthesis. During winter, deep vertical mixing distributes phytoplankton throughout a thick mixed layer while surface irradiance is low, resulting in strong light limitation. As stratification develops in spring, phytoplankton remain closer to the surface where light levels are higher, allowing growth to accelerate and initiate the seasonal bloom.

Nutrient availability provides the elemental resources required for biomass production. As phytoplankton grow during the bloom, dissolved nutrients in the upper water column are progressively consumed. When stratification reduces vertical mixing, the resupply of nutrients from deeper waters becomes limited, and nutrient limitation increasingly constrains phytoplankton growth, particularly during summer.

The realized growth rate in the model is determined by the minimum of the light and nutrient limitation terms 1.12. This formulation reflects the assumption that the scarcest resource sets the effective growth rate. Compared to a multiplicative formulation, the minimum operator avoids unrealistically strong simultaneous limitation by multiple factors and allows a clearer transition between light-limited and nutrient-limited growth regimes.

Several additional processes influence phytoplankton dynamics indirectly. Grazing by zooplankton removes phytoplankton biomass and therefore acts as a top-down control on primary producers. At high phytoplankton concentrations, the organisms themselves also increase the optical attenuation of the water column, reducing light penetration and limiting growth at depth. This self-shading mechanism introduces a negative feedback that restricts excessive biomass accumulation during bloom conditions.

Organic matter produced by primary production is transferred into the detrital pool through mortality and grazing losses. Subsequent remineralization of this detritus releases

dissolved nutrients back into the water column, providing an internal recycling pathway that can partially sustain phytoplankton production when external nutrient supply is weak.

Together, the interaction between light availability, nutrient supply, grazing pressure, self-shading, and nutrient recycling controls the seasonal evolution of phytoplankton biomass and nutrient concentrations in the model.

### 3.1.1 Limitation Functions

The limitation functions used in the model are shown in Figure 3.1. Different functional forms were chosen to represent the underlying biological and biochemical processes.

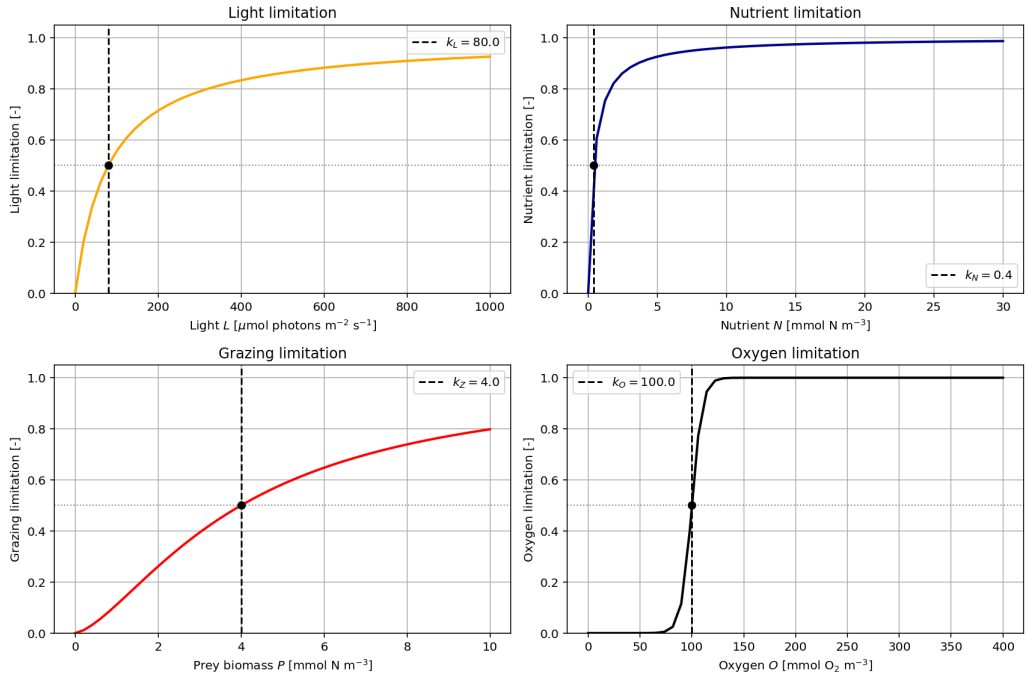


**Figure 3.1:** Functional forms of the limitation terms used in the ecosystem model.

Light and nutrient limitation are represented using classical saturation functions (Type II functional responses). These functions are widely used in ecosystem models because they capture the biochemical nature of resource uptake, where growth increases with resource availability but gradually approaches a maximum rate.

Oxygen limitation is represented using the hyperbolic tangent formulation introduced in Equation 1.17. This formulation introduces a critical oxygen threshold at which metabolic processes rapidly decline. As oxygen concentrations fall below this threshold, zooplankton activity and aerobic remineralization are strongly reduced.

The grazing limitation follows a Type III functional response, which exhibits a slower increase at low prey concentrations. This formulation reflects the ecological difficulty for

zooplankton to locate and capture prey when phytoplankton biomass is low, while allowing grazing to increase rapidly once prey becomes abundant. Such nonlinear grazing behaviour is often important for regulating bloom dynamics, as grazing pressure increases strongly once phytoplankton concentrations rise.

### 3.1.2 Vertical Structure of Limitation

To illustrate how the different limitation mechanisms interact, Figure 3.2 shows a representative vertical profile of the limitation terms together with phytoplankton biomass for a typical summer state.

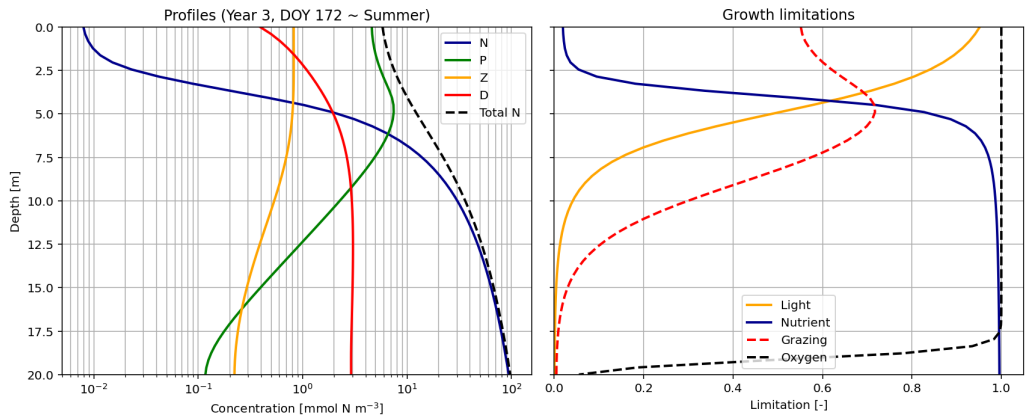


**Figure 3.2:** Vertical distribution of limitation factors and phytoplankton biomass for a representative summer state of the model.

The profile highlights the vertical structure of phytoplankton limitation in the stratified water column. Near the surface, light availability is high but nutrient concentrations are low, resulting in strong nutrient limitation. At greater depth, nutrient availability increases while light decreases, leading to increasing light limitation.

Maximum phytoplankton biomass therefore occurs at intermediate depths where both resources are sufficiently available and neither limitation dominates. This depth corresponds to the region where the light and nutrient limitation terms become comparable.

Grazing does not appear directly as a limitation curve in the figure but regulates phytoplankton biomass through zooplankton consumption. Oxygen limitation mainly becomes relevant in deeper layers, where reduced oxygen concentrations can suppress zooplankton activity and aerobic remineralization.

## 3.2 Ecosystem Response to Nutrient Loading

To investigate the effect of external nutrient supply on the ecosystem, the model was integrated under constant summer forcing until a steady-state solution was reached. Two simulations were performed representing contrasting nutrient loading scenarios: a low nutrient input case and a high nutrient input case. The resulting vertical distributions of nutrients, oxygen, phytoplankton, and detritus are shown in Figure 3.3.

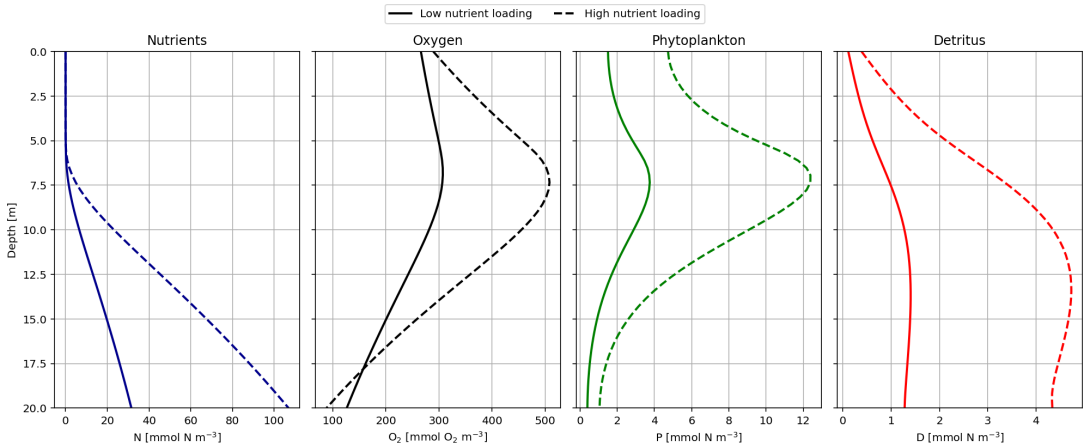


**Figure 3.3:** Steady-state vertical profiles of nutrients, oxygen, phytoplankton, and detritus for low and high nutrient loading under constant summer forcing.

In the high nutrient scenario, phytoplankton biomass is much larger at approximately the same depth as in the low nutrient case. One might initially expect the phytoplankton maximum to shift vertically. In the steady-state solution, the main change is an increase in phytoplankton concentration at the same depth, leading to higher overall biomass and stronger nutrient consumption.

The ecosystem therefore sustains more living biomass. This also produces a larger flux of organic material into the detrital pool through mortality and grazing. The higher detritus concentrations indicate that more organic matter is transported downward through the water column.

As this detrital material sinks, it is gradually remineralized by microbial processes. This remineralization converts organic nitrogen back into dissolved nutrients and consumes oxygen. As a result, nutrient concentrations increase at depth while oxygen concentrations decrease below the productive layer.

At the same time, the elevated phytoplankton biomass produces more oxygen in the productive layer around mid-depth through photosynthesis. Oxygen concentrations therefore become very high in this region. Below this layer, the increased flux of sinking detritus leads to stronger oxygen consumption and creates a much steeper oxygen gradient toward the bottom.

The gradient becomes strong enough that slightly anoxic conditions develop near the seabed. This behaviour matches real-world observations. Increased nutrient loading often leads to oxygen depletion in bottom waters, especially in shallow and stratified systems where vertical mixing is weak and oxygen resupply from the surface is limited.

## 3.3 Seasonal Ecosystem Dynamics

Seasonal forcing strongly regulates the physical environment of the water column and therefore determines how biological processes unfold over the annual cycle. The prescribed seasonal variation of vertical diffusivity, surface irradiance, and gas exchange is shown in Figure 3.4. These forcing functions control the balance between vertical transport, biological production, and oxygen consumption in the system.

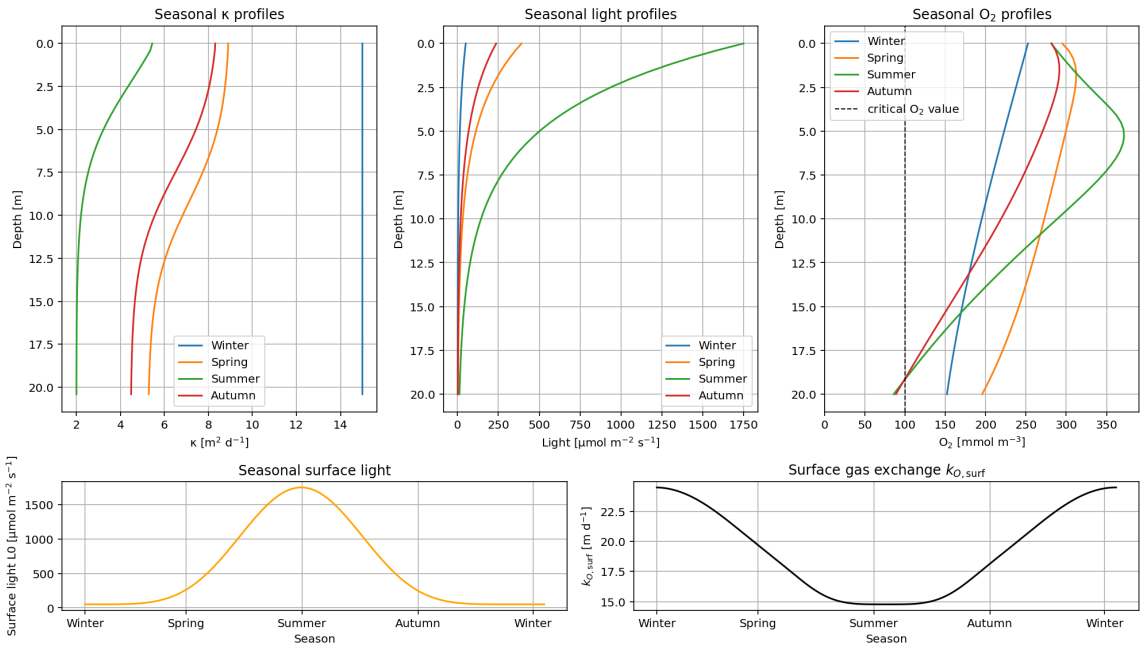


**Figure 3.4:** Seasonal forcing used in the model. Shown are the seasonal vertical diffusivity profiles, seasonal light attenuation profiles, the annual cycle of surface light intensity, and the seasonal variation of the surface gas exchange coefficient. The oxygen profiles represent the resulting seasonal oxygen distribution produced by the model rather than an externally prescribed forcing.

The dominant control on the vertical structure of the ecosystem is the seasonal variation in vertical diffusivity $\kappa$. During winter, high diffusivity values produce strong mixing that homogenizes the water column and continuously transports nutrients from depth toward the surface. Under these conditions the system remains nutrient replete but light limited. In contrast, the strong reduction in diffusivity during summer establishes stratified conditions that suppress vertical transport. Once stratification develops, nutrients in the surface layer are rapidly depleted by biological production and cannot easily be replenished

from deeper waters.

At the same time, the seasonal increase in surface irradiance initiates rapid biological growth in spring. The combination of high nutrient availability from winter mixing and increasing light allows phytoplankton production to increase abruptly, producing a pronounced bloom in the upper water column. The resulting biomass accumulation subsequently drives increases in zooplankton grazing and detritus production.

The consequences of these seasonal forcing mechanisms for the ecosystem state variables are shown in Figure 3.5. The figure illustrates the temporal and vertical evolution of nutrients, phytoplankton, zooplankton, detritus, and oxygen throughout the year under eutrophic conditions.

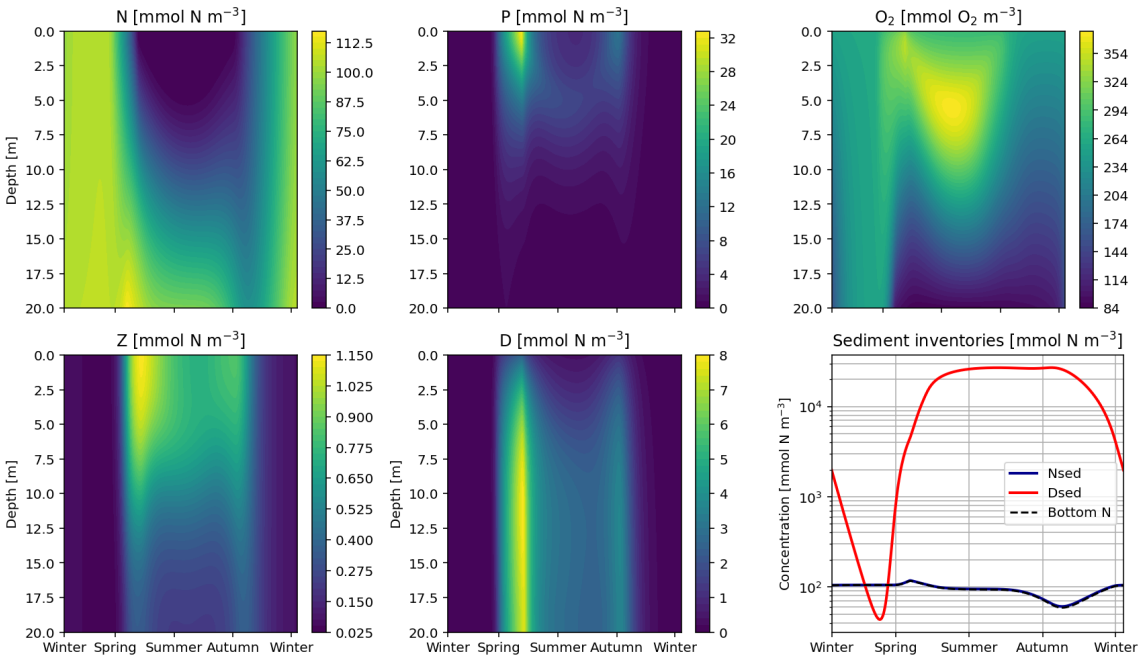


**Figure 3.5:** Seasonal evolution of ecosystem state variables. The panels show the time–depth distribution of nutrients, phytoplankton, oxygen, zooplankton, detritus, and sediment inventories over one annual cycle. The simulation represents strongly eutrophic conditions with elevated nutrient availability, which were intentionally chosen in order to investigate the development of oxygen depletion and anoxic conditions within the water column and near the seabed.

The transition from winter to spring represents the most rapid ecosystem response to the seasonal forcing. As light increases while nutrients remain abundant due to winter mixing, phytoplankton growth accelerates and generates a strong surface bloom. This bloom rapidly draws down surface nutrients and produces a large flux of organic matter through grazing and mortality.

Following the bloom, the system enters a summer regime characterized by strong stratification and weak vertical exchange. Nutrient resupply from deeper layers becomes limited, constraining further primary production. At the same time, detrital material produced

during the bloom sinks below the euphotic zone where it undergoes remineralization. This process consumes oxygen and leads to progressively lower oxygen concentrations at depth.

Because stratification simultaneously limits the downward transport of oxygen from the surface, the balance between oxygen supply and biological consumption shifts toward net oxygen loss in deeper layers. In the eutrophic simulation considered here, this process produces pronounced seasonal oxygen depletion near the seabed, demonstrating how strong primary production combined with limited vertical mixing can lead to hypoxic or anoxic conditions.

Representative seasonal vertical profiles of the biological tracers are shown in Figure 3.6. These profiles illustrate how the seasonal forcing modifies the vertical ecosystem structure over the annual cycle.

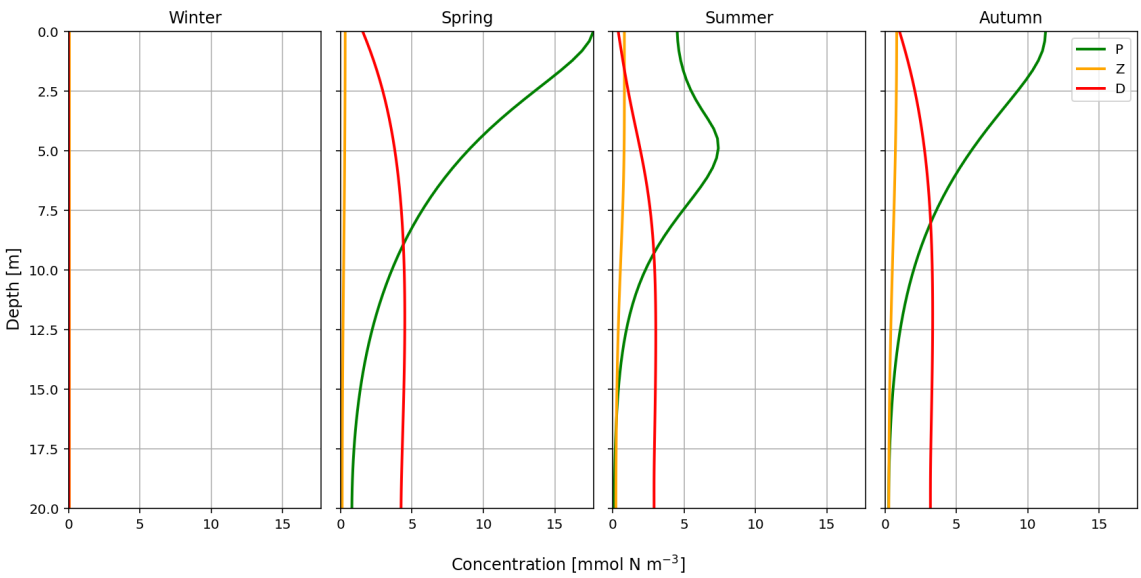


**Figure 3.6:** Representative vertical profiles of phytoplankton, zooplankton, and detritus during winter, spring, summer, and autumn.

The profiles highlight how the seasonal forcing reshapes the vertical distribution of biomass and organic matter. Winter conditions maintain uniformly low biomass due to light limitation. In spring, production becomes concentrated near the surface, generating a pronounced phytoplankton maximum. During summer, biomass declines in the nutrient-depleted surface layer while detrital material accumulates below the productive zone due to sinking export. Finally, autumn mixing partially restores nutrients in the upper water column, allowing a modest recovery of phytoplankton biomass before winter conditions return.

Overall, the seasonal forcing primarily acts by modulating vertical mixing and light availability. These physical controls determine the timing of nutrient resupply, the development of phytoplankton blooms, the export of organic matter, and ultimately the seasonal formation of oxygen deficits in the deeper water column.

## 3.4 Sensitivity Analysis

To assess how strongly the simulated ecosystem depends on parameter choices, a set of sensitivity experiments was performed. In these simulations, selected biological parameters were varied while all other parameters and forcing conditions were kept identical. The resulting steady-state vertical profiles were compared to the reference simulation.

Three parameters were examined: phytoplankton mortality, the nutrient half-saturation constant for phytoplankton uptake, and the sinking velocity of detrital particles. These parameters influence different parts of the ecosystem: biomass loss, nutrient uptake efficiency, and the vertical transport of organic matter.

### 3.4.1 Sensitivity to Phytoplankton Mortality

The first experiment investigates the influence of the phytoplankton mortality rate $m_P$. Figure 3.7 compares the resulting steady-state profiles.

$$P \xrightarrow{m_P} D \xrightarrow{r} N, \tag{3.1}$$

Increasing the mortality rate reduces phytoplankton biomass throughout the water column because biomass is removed more rapidly. The additional detritus production enhances remineralization and slightly increases nutrient concentrations. The phytoplankton maximum also shifts slightly toward shallower depths, where light conditions allow growth to balance the stronger losses.

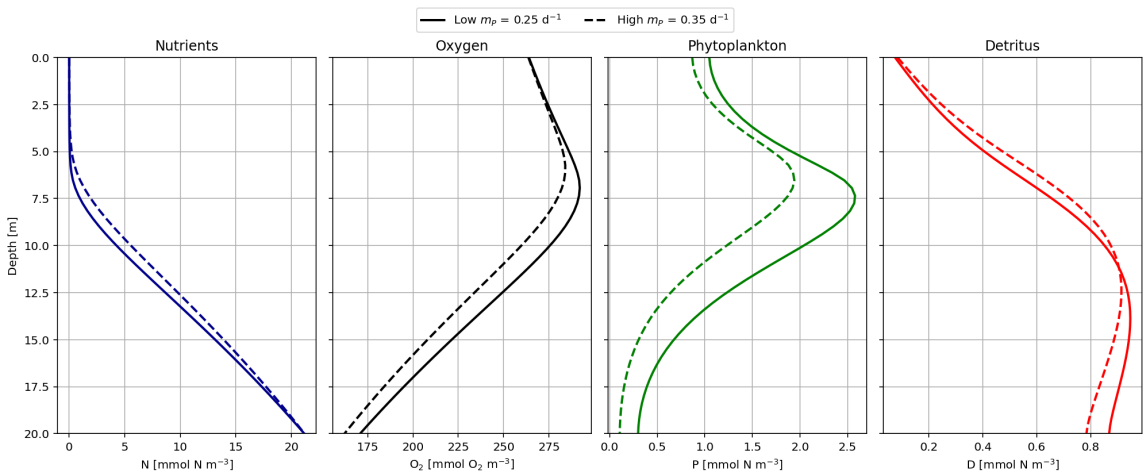


**Figure 3.7:** Sensitivity of the ecosystem structure to changes in phytoplankton mortality rate.

Higher detritus production increases remineralization below the productive layer, which leads to slightly stronger oxygen depletion at intermediate depths. One possible real-world

analogue of this mechanism could be increased biological mortality associated with rising temperatures under climate change. If such effects become significant, ecosystems may exhibit reduced phytoplankton biomass and a slightly shallower biomass maximum due to stronger loss processes.

### 3.4.2 Sensitivity to Nutrient Uptake Efficiency

The second experiment examines the nutrient half-saturation constant $k_N$, which controls the efficiency of nutrient uptake.

$$\mu = \mu_{\max} \min\left(\frac{L}{L+k_L}, \frac{N}{N+k_N}\right), \tag{3.2}$$

$$k_N \uparrow \Rightarrow \; P \downarrow \Rightarrow \; \text{nutrient uptake} \downarrow . \tag{3.3}$$

Increasing $k_N$ slightly reduces phytoplankton biomass because growth becomes more strongly nutrient limited. As a result, nutrient concentrations remain somewhat higher. The overall vertical structure of the ecosystem changes only weakly, and the position of the phytoplankton maximum remains nearly unchanged, because the location of the biomass maximum is primarily controlled by the balance between light and nutrient availability rather than by moderate changes in uptake efficiency.

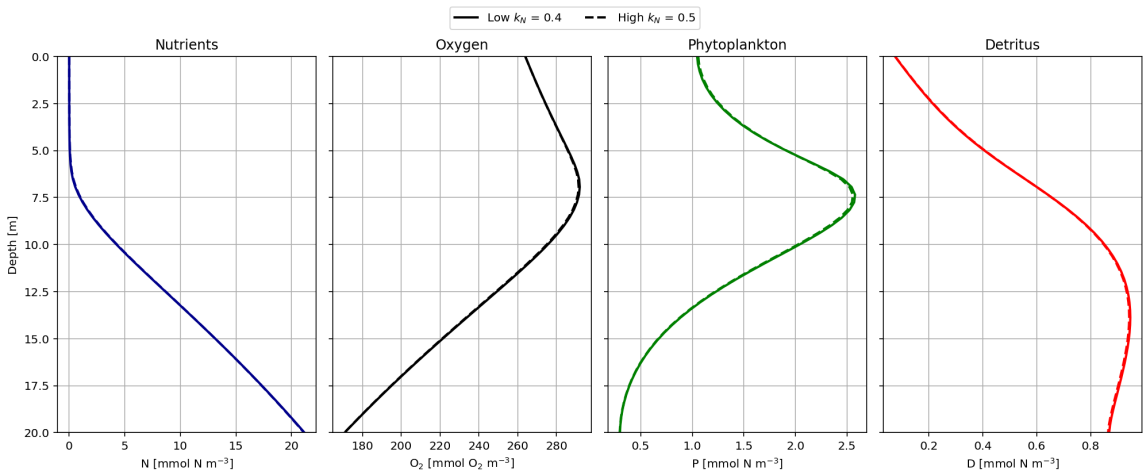


**Figure 3.8:** Sensitivity of the ecosystem structure to changes in the nutrient half-saturation constant.

### 3.4.3 Sensitivity to Detritus Sinking Velocity

The final experiment explores the effect of the detritus sinking velocity $w_D$, which controls the vertical export of organic matter.

$$\frac{\partial D}{\partial t} = -w_D \frac{\partial D}{\partial z} + \text{sources} - \text{sinks}, \tag{3.4}$$

$$w_D \uparrow \Rightarrow \text{ detritus export} \uparrow \Rightarrow \text{ remineralization at depth} \uparrow . \tag{3.5}$$

Increasing the sinking velocity transports detritus more rapidly toward deeper layers. Detritus concentrations therefore decrease in the upper water column, while remineralization and nutrient concentrations increase at depth. The vertical position of the phytoplankton maximum remains largely unchanged because growth is primarily controlled by the balance between light and nutrient limitation. Its magnitude decreases slightly, as less remineralization occurs near the phytoplankton layer, reducing the local recycling of nutrients.

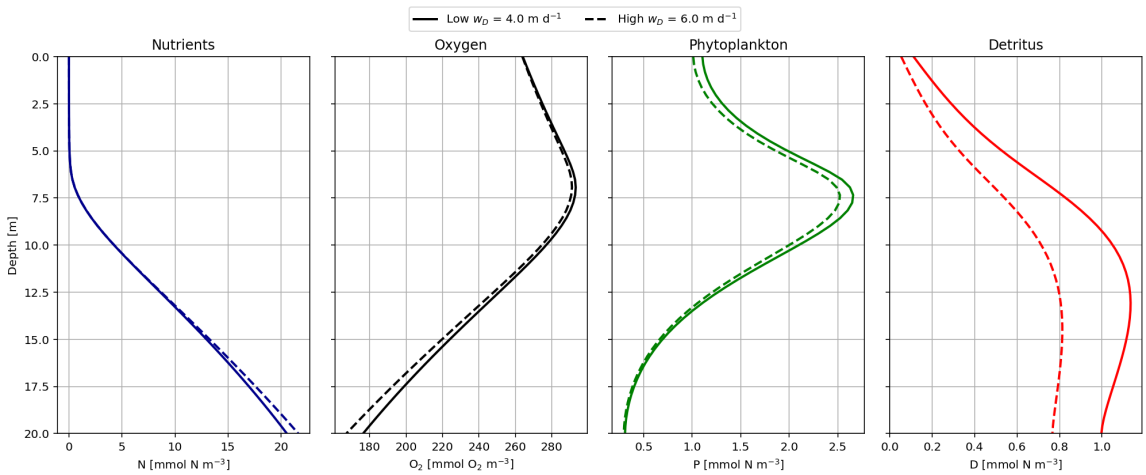


**Figure 3.9:** Sensitivity of the ecosystem structure to changes in detritus sinking velocity.

This behaviour can be interpreted as a simplified representation of differences between plankton communities. Ecosystems dominated by larger phytoplankton species, such as diatoms, typically produce heavier particles that sink faster and therefore transport organic matter more efficiently to depth. In contrast, systems dominated by smaller phytoplankton tend to produce slower-sinking particles, resulting in stronger recycling of nutrients within the upper water column.

# Conclusion

The ecosystem model developed in this study provides a simplified representation of nutrient cycling, biological production, and oxygen dynamics in a vertically resolved water column. Despite its conceptual simplicity, the model reproduces several key features of seasonal marine ecosystems. In particular, the simulations capture the typical succession of winter nutrient accumulation, a pronounced spring phytoplankton bloom, summer nutrient limitation, and oxygen depletion below the productive surface layer. The timing of the spring bloom and the general magnitude of phytoplankton biomass emerge naturally from the seasonal variations in light availability and vertical mixing, demonstrating that these physical drivers exert a strong first-order control on ecosystem dynamics.

The model further illustrates how biological production in the surface layer is linked to oxygen consumption at depth through sinking organic matter and subsequent remineralization. Sensitivity experiments highlight the importance of parameters controlling phytoplankton losses and particle export, which regulate the partitioning between living biomass and detritus and therefore influence subsurface oxygen dynamics.

At the same time, the simplified formulation imposes clear limitations. The model only considers nitrogen as the limiting nutrient and does not represent other potentially important elements such as phosphorus or sulfur, which can play a crucial role in highly productive or oxygen-depleted environments. In addition, the sediment is represented as a depth-integrated reservoir, which does not resolve the internal biogeochemical structure of benthic sediments.

Future work could therefore extend the model by introducing multiple phytoplankton functional types with different ecological traits. Distinguishing between groups such as small phytoplankton and larger, faster-sinking species would allow the model to represent differences in nutrient uptake efficiency, growth strategies, and particle export. Such an extension would improve the representation of ecosystem structure and could help explain how shifts in plankton community composition influence vertical export of organic matter and the development of oxygen depletion at depth. Nevertheless, the present model already demonstrates how a relatively simple ecosystem formulation can reproduce the main seasonal patterns of marine productivity and oxygen variability.